WELCOME TO
THE CORE PILLARS OF INDUSTRY 4.0

In this session, we will explore the key technology pillars that power Industry 4.0 and enable intelligent, connected and future-ready enterprises.

SESSION AGENDA

- 01 Internet of Things (IoT)
- 02 Artificial Intelligence (AI)
- 03 Cloud Computing
- 04 Big Data Analytics
- 05 Blockchain
- 06 Cybersecurity
- 07 Digital Twins
- 08 Edge Computing

KEY MESSAGE: These technologies work together to create **intelligent**, **agile** and **sustainable** enterprises in the era of Industry 4.0.

INTERNET OF THINGS (IOT):
THE EYES AND EARS

IoT connects physical devices and sensors to collect, exchange and transmit data in real time.

WHAT IS IOT?

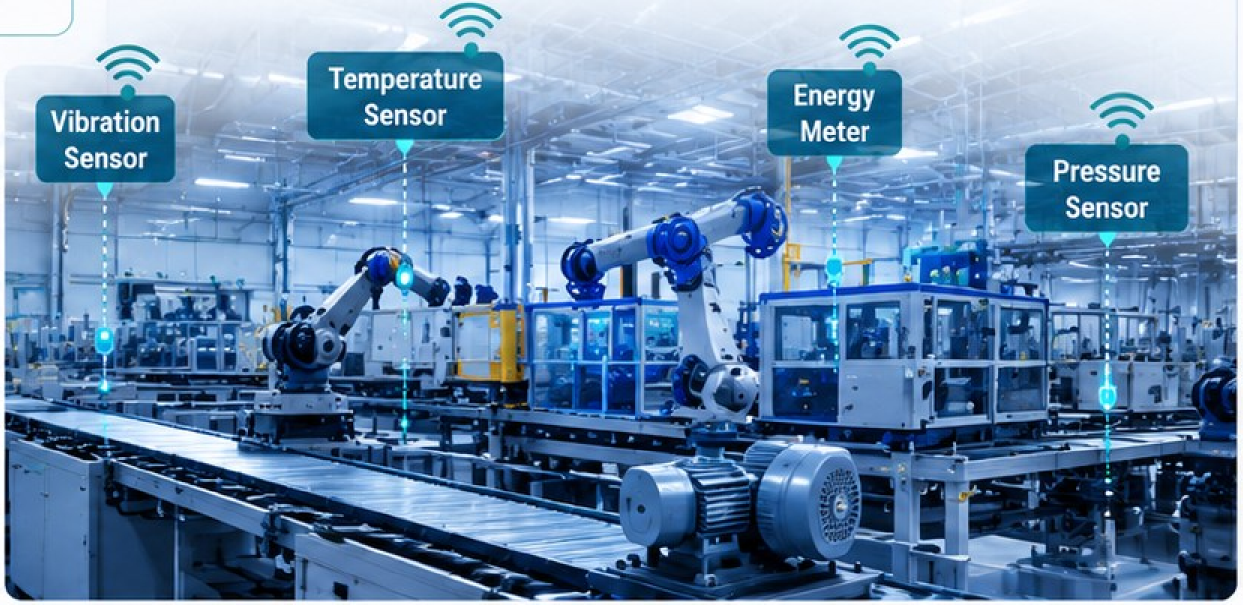
A network of connected devices and sensors that collect, share and communicate data over the internet.

KEY FUNCTIONS

- Data Collection
- Real-Time Monitoring
- Asset Tracking
- Remote Control
- Predictive Maintenance
- Process Optimization

HOW IOT WORKS

Sensors / Devices → Connectivity (Wi-Fi / 5G / LoRa) → IoT Platform (Cloud / Edge) → Applications & Dashboards



EXAMPLE IN ACTION

A machine equipped with vibration and temperature sensors sends real-time data to the IoT platform.

The system monitors the health of the machine and alerts the team if something goes wrong.

This helps in reducing downtime and extending equipment life.

Sensor → Data → Alert

KEY MESSAGE

IoT is the foundation of Industry 4.0.
It captures the real-world data that powers intelligent decisions.

BUSINESS IMPACT

- Reduced Downtime
- Lower Maintenance Costs
- Improved Safety
- Higher Efficiency
- Better Decision Making

ARTIFICIAL INTELLIGENCE (AI):
THE BRAIN

AI turns data into intelligence by learning patterns, making predictions and automating decisions to optimize operations.

WHAT IS AI?

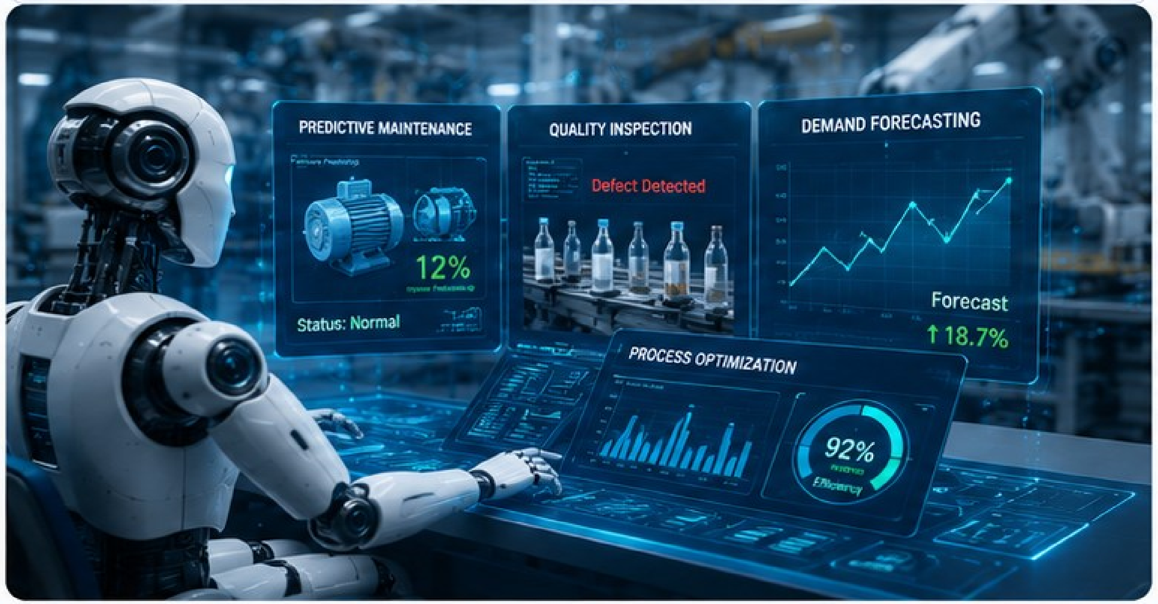
AI refers to systems and algorithms that can learn from data, reason, predict outcomes and make intelligent decisions with minimal human intervention.

KEY CAPABILITIES

- Machine Learning
- Predictive Analytics
- Anomaly Detection
- Computer Vision
- Natural Language Processing
- Optimization & Decision Making
- Automation & Autonomy

HOW AI WORKS

Data from IoT Devices & Systems → AI/ML Algorithms Process the Data → Insights & Predictions → Intelligent Actions & Automation



EXAMPLE IN ACTION

AI analyzes vibration and temperature data from machines to predict possible failures before they occur.

It also detects defects in products using computer vision and recommends the best course of action automatically.

This helps reduce downtime, improve quality and maximize productivity.

Data → AI Analysis → Prediction → Action

KEY MESSAGE

AI is the **brain of Industry 4.0.**
It analyzes complex data, learns continuously and enables smarter, faster and more accurate decisions.

BUSINESS IMPACT

- Predicts Failures Before They Happen
- Improves Quality & Consistency
- Optimizes Processes & Resources
- Enhances Customer Experience
- Drives Innovation & Competitive Advant

CLOUD COMPUTING & BIG DATA ANALYTICS:

POWERING SCALABILITY & INSIGHTS

Cloud scales your data.
Big Data turns it into smarter decisions.

HOW IT WORKS TOGETHER

Collect Data → Store & Process → Analyze → Visualize → Act

CLOUD COMPUTING

- Scalable
- Reliable & Available
- Global Access
- Secure & Cost Efficient

BIG DATA ANALYTICS

- Ingest Data
- Process Data
- Analyze Data
- Generate Insights

EXAMPLE IN ACTION

- Sensors generate data.
- Cloud stores and processes it.
- Analytics delivers insights.

KEY MESSAGE

Cloud provides **scale**. Analytics delivers **intelligence**.

BUSINESS IMPACT

- Lower Costs
- Faster Insights
- Better Efficiency
- Reduced Risk
- Better Customer Experience
- Data-Driven Growth

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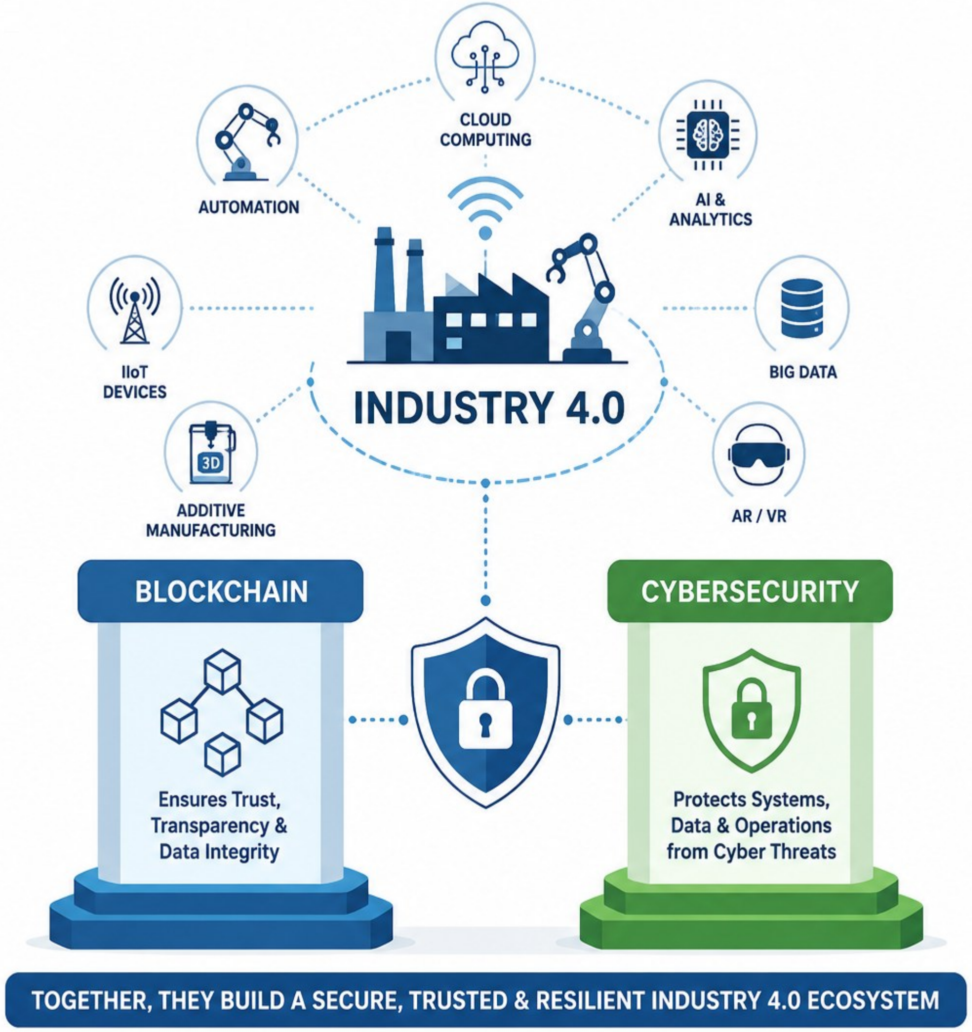
BLOCKCHAIN & CYBERSECURITY:
TRUST, TRANSPARENCY & PROTECTION

Building a secure, trustworthy and resilient digital foundation for Industry 4.0

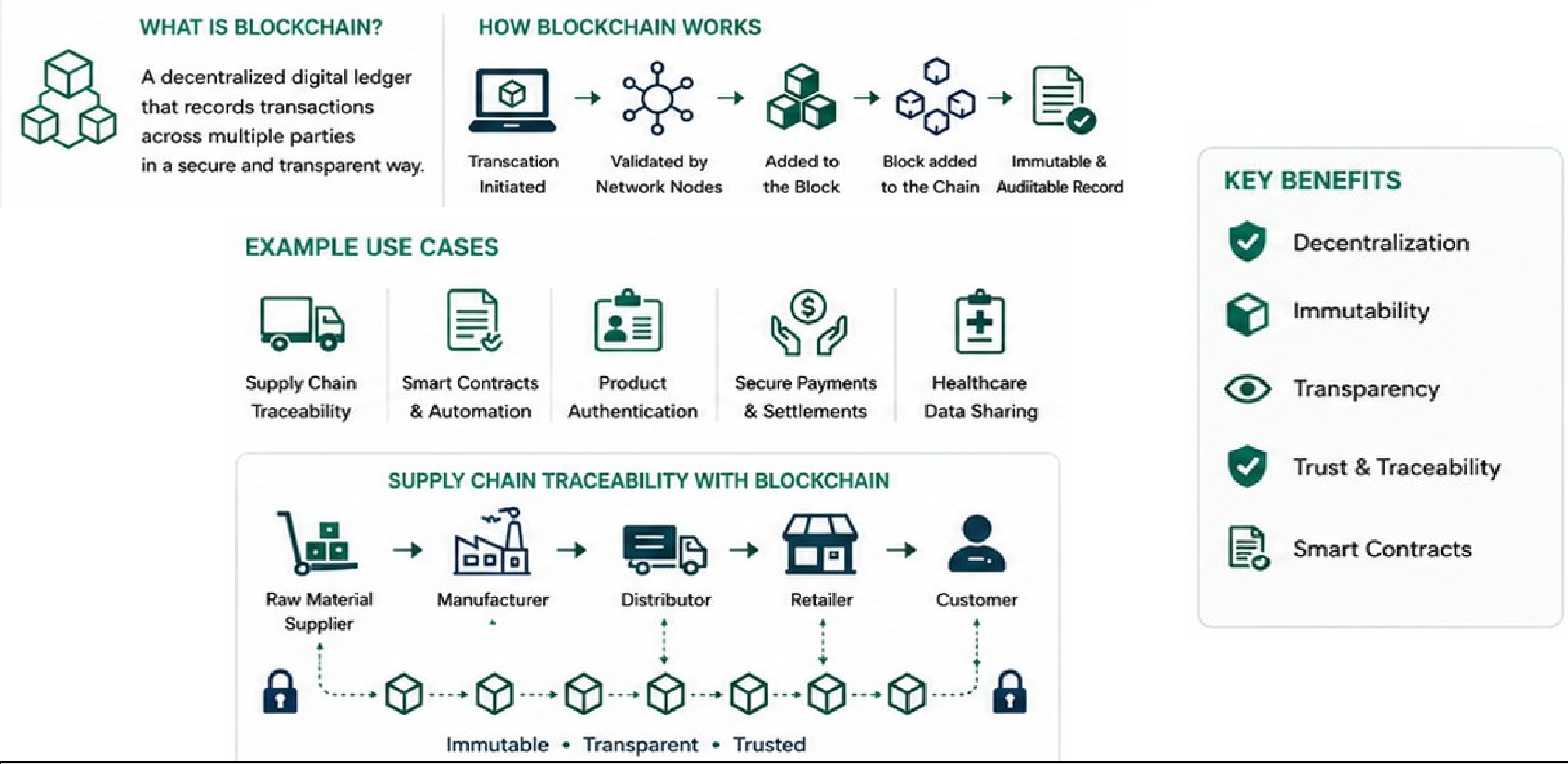
Industry 4.0 – Hyper connected with billions of smart devices, industrial systems, entire organizations, and global supply chains.

- **First, the question of Trust:** In a vast network where data changes hands millions of times a second, *how can we verify that the information being shared is authentic, untampered, and accurate?*

- **Second, the question of Security:** As our digital attack surface grows exponentially, *how do we shield this critical infrastructure from sophisticated cyber threats?*



BLOCKCHAIN : THE TRUST LAYER



CYBERSECURITY : THE PROTECTION LAYER

WHAT IS CYBERSECURITY?

The practice of protecting systems, networks, devices and data from unauthorized access, attacks and damage.

CORE SECURITY OBJECTIVES

CONFIDENTIALITY
Ensure data is accessible only to authorized users.

INTEGRITY
Ensure data is accurate, consistent and tamper-free.

AVAILABILITY
Ensure systems and data are available when needed.

KEY SECURITY CONTROLS

Identity & Access Management
Control who can access what

Network Security
Protect networks and communication

Data Encryption
Encrypt data in transit and at rest

Threat Detection & Monitoring
Detect, analyze and respond to threats

Security Policies & Compliance
Follow standards and regulatory requirements

EXAMPLE IN ACTION

- Protects against cyber threats and attacks
- Ensures safe and continuous operations
- Builds trust and safeguards business reputation

BLOCKCHAIN & CYBERSECURITY:
TRUST, TRANSPARENCY & PROTECTION

Building a secure, trustworthy and resilient digital foundation for Industry 4.0

BLOCKCHAIN: THE TRUST LAYER

WHAT IS BLOCKCHAIN?

A decentralized digital ledger that records transactions across multiple parties in a secure and transparent way.

HOW BLOCKCHAIN WORKS

Transcation Initiated → Validated by Network Nodes → Added to the Block → Block added to the Chain → Immutable & Auditable Record

KEY BENEFITS

- Decentralization
- Immutability
- Transparency
- Trust & Traceability
- Smart Contracts

EXAMPLE USE CASES

- Supply Chain Traceability
- Smart Contracts & Automation
- Product Authentication
- Secure Payments & Settlements
- Healthcare Data Sharing

SUPPLY CHAIN TRACEABILITY WITH BLOCKCHAIN

Raw Material Supplier → Manufacturer → Distributor → Retailer → Customer

Immutable • Transparent • Trusted

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EXAMPLE IN ACTION

Connected Devices ↔ Secure Network & Gateway ↔ Cloud / Data Center; Secure Applications & Dashboards

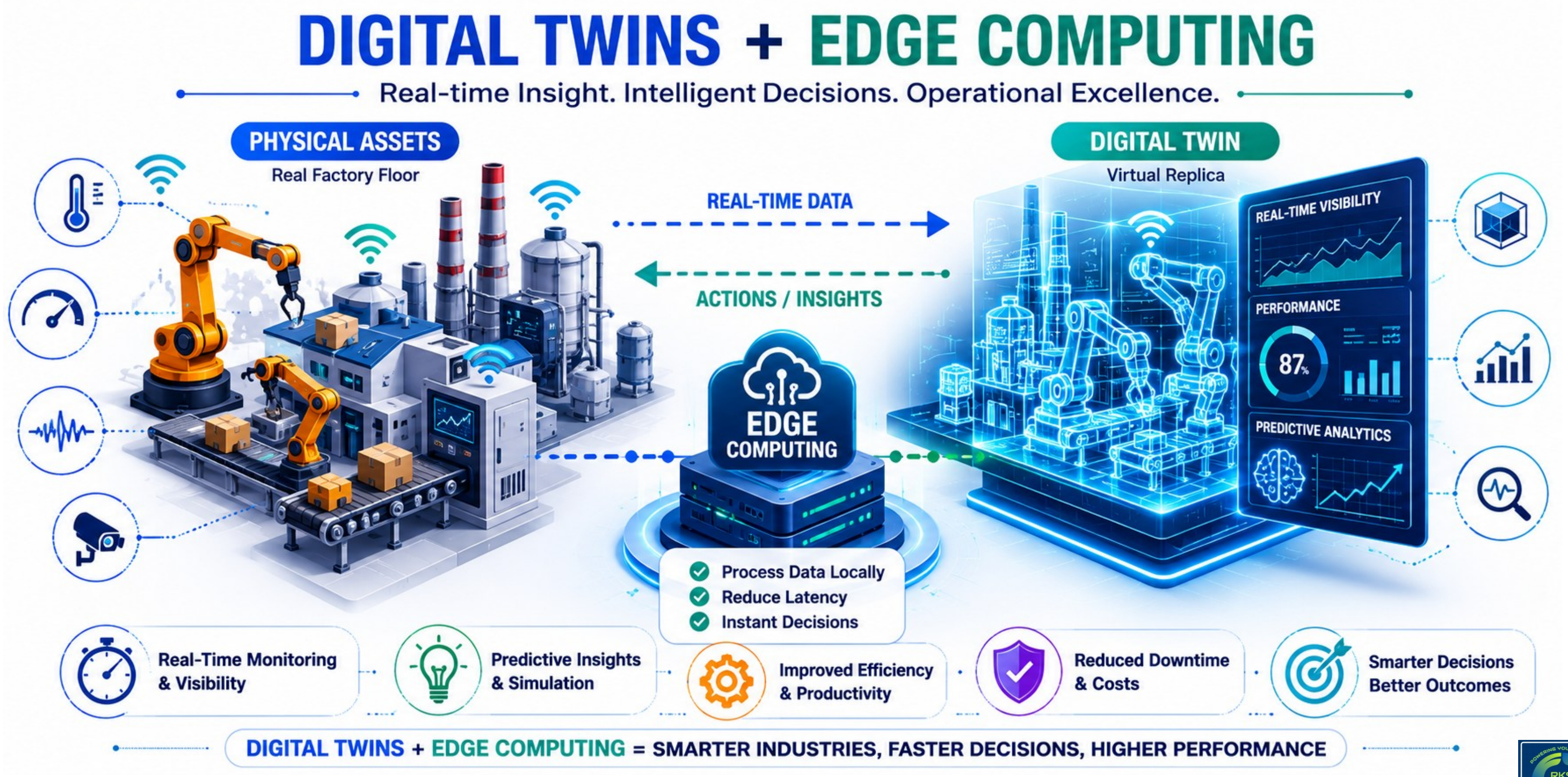
- ✓ Protects against cyber threats and attacks
- ✓ Ensures safe and continuous operations
- ✓ Builds trust and safeguards business reputation

BUSINESS IMPACT

- Stronger Trust & Transparency
- Reduced Risks & Cyber Threats
- Operational Resilience
- Regulatory Compliance & Audit Readiness
- Better Collaboration Across Ecosystem
- Secure Data, Stronger Business

DIGITAL TWINS & EDGE COMPUTING:
REAL-TIME INTELLIGENCE, CLOSER TO ACTION

Enabling real-time simulation, monitoring and decision-making at the edge of the network.



DIGITAL TWIN : THE VIRTUAL REPRESENTATION

WHAT IS DIGITAL TWIN?

A virtual replica of a physical asset, process or system that uses real-time data to simulate, monitor, analyze and optimize performance.

HOW DIGITAL TWIN WORKS

Real World Asset → Data from IoT Sensors → Digital Twin Model → Simulation & Analytics → Optimized Decisions

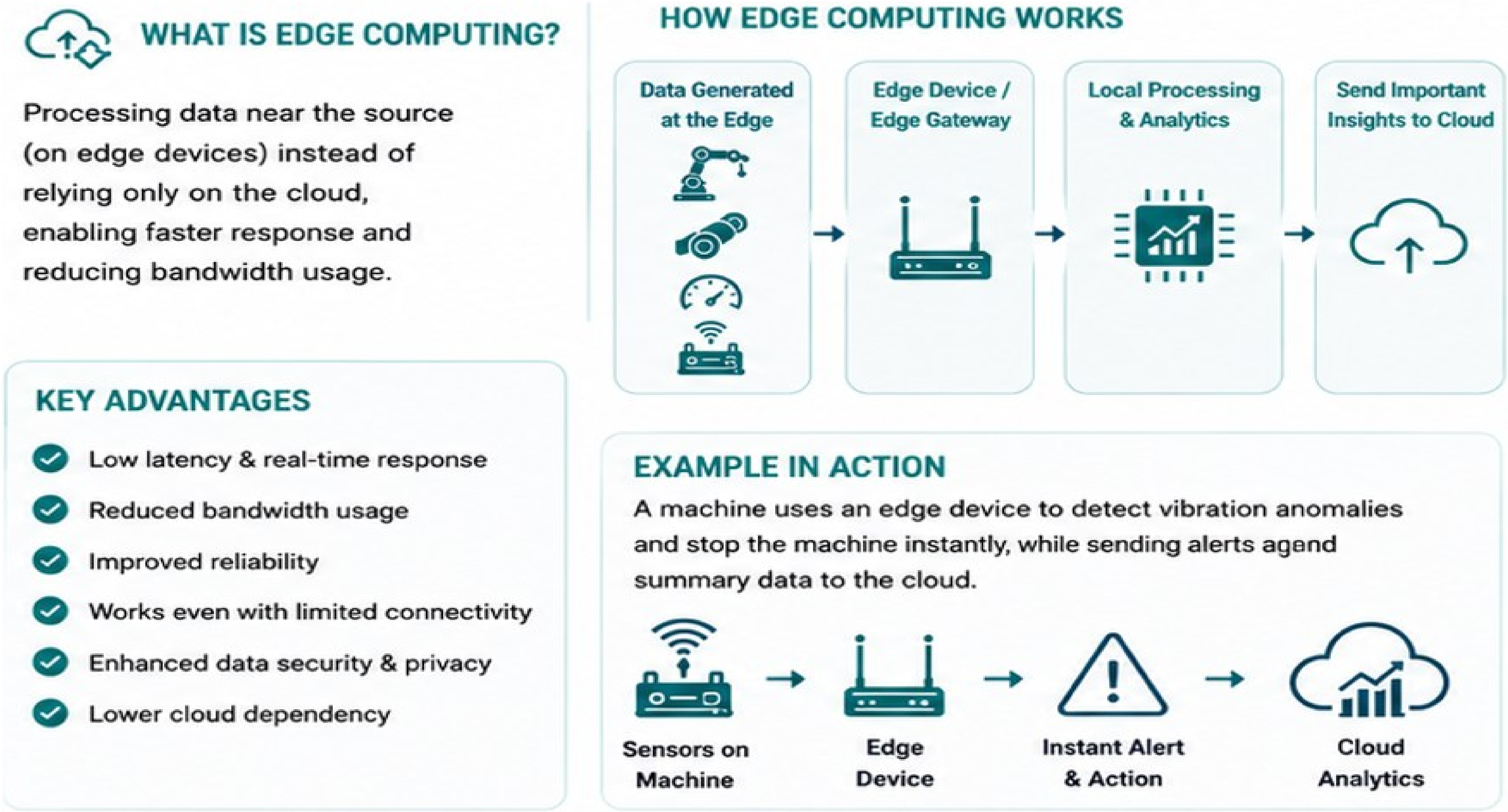
KEY BENEFITS

- Better design & validation
- Real-time monitoring
- Predictive maintenance
- Process optimization
- Reduced downtime & cost
- What-if scenario analysis
- Improved product quality

EXAMPLE IN ACTION

A digital twin of a production line simulates different conditions, predicts failures and recommenlds the best actions to maximize performance and reduce downtime.

EDGE COMPUTING : REAL TIME PROCESSING LAYER



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EDGE COMPUTING: THE REAL-TIME PROCESSING LAYER

WHAT IS EDGE COMPUTING?

Processing data near the source (on edge devices) instead of relying only on the cloud, enabling faster response and reducing bandwidth usage.

HOW EDGE COMPUTING WORKS

Data Generated at the Edge → Edge Device / Edge Gateway → Local Processing & Analytics → Send Important Insights to Cloud

KEY ADVANTAGES

- Low latency & real-time response
- Reduced bandwidth usage
- Improved reliability
- Works even with limited connectivity
- Enhanced data security & privacy
- Lower cloud dependency

EXAMPLE IN ACTION

A machine uses an edge device to detect vibration anomalies and stop the machine instantly, while sending alerts agend summary data to the cloud.

Sensors on Machine → Edge Device → Instant Alert & Action → Cloud Analytics

BUSINESS IMPACT

- Faster Decision Making
- Lower Costs & Operational Risk
- Higher Efficiency & Productivity
- Greater Reliability & Resilience
- Better Quality & Customer Satisfaction
- Innovation & Competitive Advantage

THE INDUSTRY 4.0 TECHNOLOGY STACK

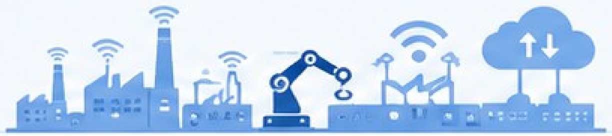
A layered architecture showing how technologies build on each other to deliver value and drive intelligent operations

KEY IDEA
Each layer depends on the one below it. Together, they enable end-to-end intelligence and automation.

Layer	Components	Role
5. BUSINESS APPLICATIONS	Dashboards & Visualization; Operations Management; Customer Experience; Asset & Service Management; Strategic Decision Making	Delivers business value through intelligent insights, automation and optimized processes.
4. ANALYTICS & AI	Data Analytics; Machine Learning; Predictive Models; Optimization; Automation & Decisioning	Transforms data into intelligence using analytics, AI and machine learning to predict, optimize and automate.
3. CLOUD PLATFORM	Storage; Compute; Database; Middleware Services; APIs & Integration	Provides scalable infrastructure, storage, processing power and integration services.
2. CONNECTIVITY LAYER	Wi-Fi; 5G / LTE; Ethernet; Bluetooth; LoRa / NB-IoT	Ensures reliable and secure communication between devices, systems and the cloud.
1. DEVICES & SENSORS	Sensors; Actuators; Machines; PLC / SCADA; Smart Devices	Captures real-world data and performs actions in the physical environment.

KEY MESSAGE

Industry 4.0 is built as a connected technology ecosystem.
Strong foundations at the bottom enable smarter outcomes at the top.



BRINGING IT ALL TOGETHER:
THE INDUSTRY 4.0 ECOSYSTEM

Integrated technologies. Intelligent operations. Transformative outcomes.

ONE CONNECTED ECOSYSTEM

Data flows across the value chain.
Intelligence drives decisions. Automation creates value.

END-TO-END DATA & VALUE FLOW

1. **SENSORS & DEVICES** — Capture real-time data.
2. **INTERNET OF THINGS (IOT)** — Connects devices and transmits data.
3. **EDGE COMPUTING** — Process data locally for real-time insights and actions.
4. **CLOUD PLATFORM** — Store, manage and secure data at scale.
5. **BIG DATA ANALYTICS** — Identify patterns and generate insights.
6. **ARTIFICIAL INTELLIGENCE** — AI learns, predicts and automates decisions.
7. **BLOCKCHAIN & CYBERSECURITY** — Ensure trust, security and traceability.
8. **BUSINESS OUTCOMES** — Insights drive better decisions and business results.

Act
Monitor
Optimize
Innovate

CONTINUOUS IMPROVEMENT

Outcomes create a continuous loop of learning, optimization and innovation.

CROSS-CUTTING FOUNDATIONS

- Cybersecurity & Privacy
- Connectivity & Networks
- Data Governance & Quality
- People, Skills & Culture
- Standards & Interoperability

BUSINESS IMPACT

- Higher Efficiency
- Lower Downtime & Costs
- Better Quality & Compliance
- Enhanced Customer Experience
- New Revenue Streams
- Sustainable & Resilient Operations

KEY TAKEAWAY

Industry 4.0 happens when data, intelligence, trust and automation come together to create smarter, faster and more resilient enterprises.

Connect. Analyze. Automate. Innovate.
That's the Power of Industry 4.0.